1. String Concatenation

Write a Python program that takes two strings i.e string 1 “Hello ”, string 2 get name as input from the user and concatenates them together. Display the concatenated string as the output.

```
string1 = "Hello "
string2 = input("Enter your name: ")
print(string1+string2)
```

```
Enter your name: GOPAL V
Hello GOPAL V
```

Now concatenate string 3 “, welcome to Python programming” to the existing string and display the output string.

```
string3 = ", Welcome to Python programming"
greet = string1+string2+string3
print(greet)
```

```
Hello GOPAL V, Welcome to Python programming
```

2. String Slicing and Indexing:

Write a Python program using the above concatenated string as input and performs the following tasks:

```
# a. Print the first character of the string.
print('First character: ', greet[0])
```

```
First character:  H
```

```
# b. Print the last character of the string.
print('Last character: ', greet[-1])
```

```
Last character:  g
```

```
# c. Print the first 5 characters of the string.
print('First 5 characters:', greet[0:5])
```

```
First 5 characters: Hello
```

```
# d. Print the last 11 characters of the string.
print('Last 11 characters:', greet[-11:])
```

```
Last 11 characters: programming
```

```
# e. Print the string in reverse.
print('Reverse: ', greet[::-1])
```

```
Reverse:  gnimmargorp nohtyP ot emocleW ,V LAPOG olleH
```

```
# f. Use slicing and print the word “Python” from the existing string.
print(greet[-18:-12])
```

```
Python
```

3. String Methods:

Write a Python program that takes a string, strM = “Python beginner tutorial” and perform the following tasks:

```
# a. Convert the sentence to uppercase.
strM = "Python beginner tutorial"
print(strM.upper())
```

```
PYTHON BEGINNER TUTORIAL
```

```
# b. Convert the sentence to lowercase.
print(strM.lower())
```

```
python beginner tutorial
```

```
# c. Use Capitalize and return the sentence to the original input form.
print(strM.capitalize())
```

```
Python beginner tutorial
```

```
# d. Count the total number of occurrences of character ‘t’ in the string.
print(strM.count('t'))
```

```
3
```

```
# e. Replace all occurrences of “Python” with “Machine Learning” in the input string strM = “Python beginner tutorial”
print(strM.replace('Python','Machine Learning'))
```

```
Machine Learning beginner tutorial
```

Tuples

Create 1st tuple with values -> (10, 20, 30), 2nd tuple with values -> (40, 50, 60):

```
# a. Concatenate the two tuples and store it in “t_combine”
t1 = (10,20,30)
t2 = (40,50,60)
t_combine = t1+t2
print(t_combine)
```

```
(10, 20, 30, 40, 50, 60)
```

```
# b. Repeat the elements of “t_combine” 3 times
print(t_combine * 3)
```

```
(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
```

```
# c. Access the 3rd element from “t_combine”
print('3rd element: ',t_combine[2])
```

```
3rd element:  30
```

```
# d. Access the first three elements from “t_combine”
print('first three elements: ',t_combine[0:3])
```

```
first three elements:  (10, 20, 30)
```

```
# e. Access the last three elements from “t_combine”
print('last three elements: ',t_combine[-3:])
```

```
last three elements:  (40, 50, 60)
```